

1.1.1 Simple introduction to the Internet itself

Explains today's communication as a multiple protocols.

Introduction of RFC - ARPANET Reference number.

It's the description of Internet protocols architecture

MMU

This part is about primary goal of internet:

"develop an effective technique or multiplexed utilization of existing multiconnected networks"

In Leymon's terms the main goal of Internet is connecting multiple existing local networks to enable them to connect to each other.

Also author introduces some secondary goals that will be described later.

1.1.1 Packet connection

Today's network ~~was~~ was created according to the telephone network. Although some changes was implemented.

So all the data is sending in "chunks" what means that many small elements are traveling through independently. That design brings two major advantages: resilience and better utilization of the bandwidth.

Last theoretical detail is the the packets of many senders can mix up and be put apart later what create Multiplexing

1.1.2 End to end and Fate sharing

(E2E) this principle states that advance features like security or reliability should be done by the end point of communication then by the network layer itself. This attitude introduces

"dumb" network and "smart" systems connected to the network

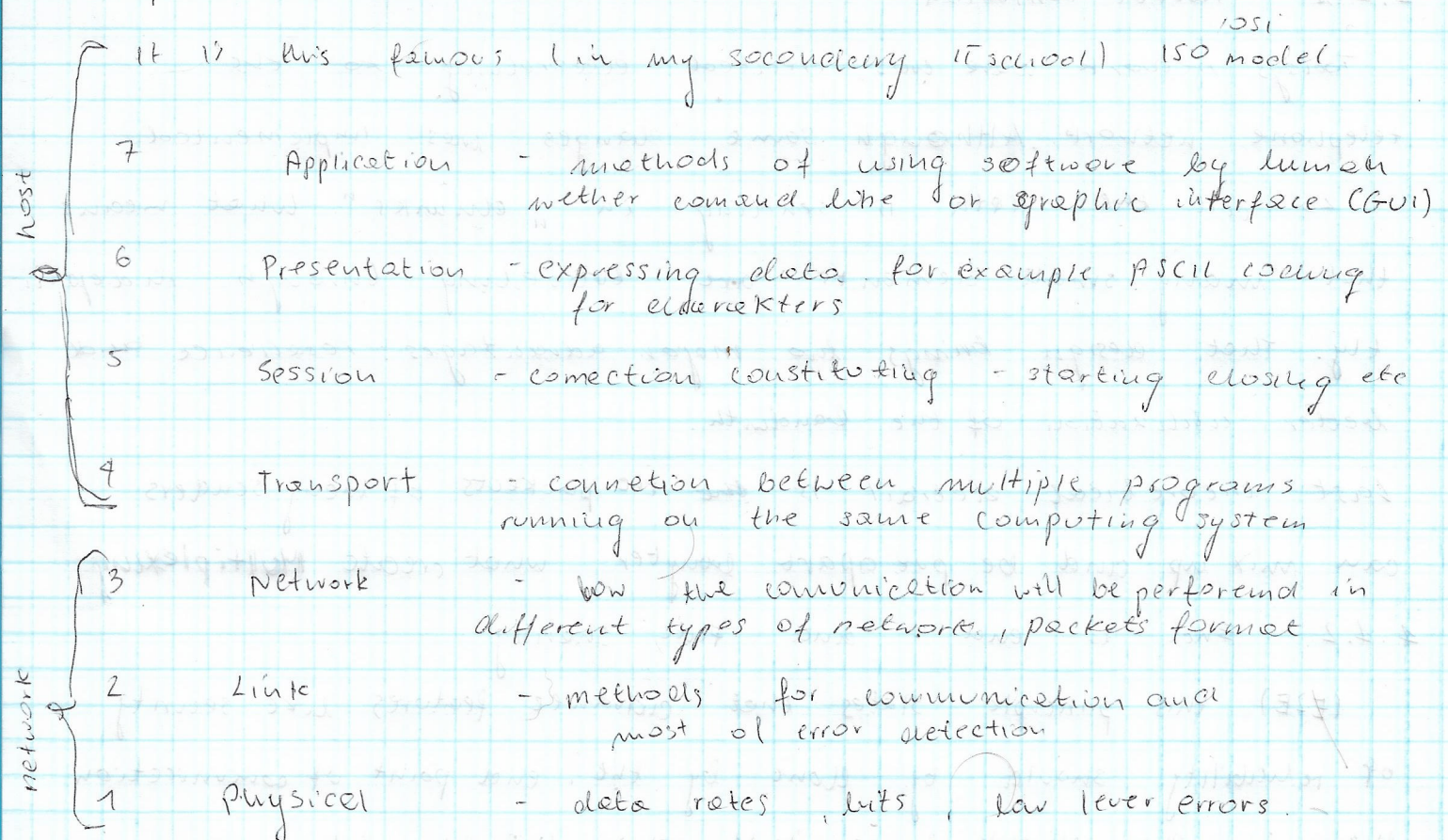
Fate sharing is a usage of E2E design that require critical session data to be kept by hosts instead of the network

1.1.3. error and flow control

short paragraph about errors during chunks sending
sometimes it is required to deliver chunks in exact order
so in case of troubles uses flow control to slowdown
sending packets to resend corrupted packets and keep
the order. But mostly the order is not obligatory what
speeds up the speed of transfer! corrupted packets
can be send even ok the and so the speed does not
slow down. Only time lost is the time that corrupted packets
require to be resend).

1.2 / 1.2.1

layering - split the complex implementation into smaller
chunks - layers in this case - that are easier to manage and
implement.



Multiplexing, Demultiplexing and encapsulation

in simple terms building object is similar to assembling the train. every carriage has passengers same like every element has its own data. When we are assembling the package we are combining multiple carriages where passengers don't know each other between carriages. Mostly following layers used to add data at the beginning, but nowadays it uses to add data either at the beginning and at the end of packet.

encapsulation - is a stripping layers belonging to the protocols when package is "reading"

Protocol data unit - is the data belonging to the specified protocol in a package.

1.3. TCP/IP suite

7 Application

4 Transport

3.5 Network
(Address)

3 Network - IP4 / IPv6 addressing

2.5 Link - mapping addresses like ARP

forwarding:

unicast - for a single host

broadcast - for all hosts in specific network.

multicast - for set of hosts belonging to multicast group

also some other protocols are introduced there but they'll be discussed later in detail so I skip them for now.

Port numbers - 16-bit int every IP number has a 65,536 ways to forward the data to the transport protocols it works a little bit like IP extension, what means you can call to the specific person in company instead of calling the general number and asking for specific person

3 port ranger

1. well known (0, 1023) for protocols, 2. registered (1024-49151)
3. dynamic private (49152-65536)

1.4. Internet, internet, Intranet and extranet

Internet - aggregate network resulting from the interconnection of constituent networks over time
(^{of} ~~internet~~ ^{interconnection})

Internet - collection of hosts around the world that can connect with each other using TCP/IP

Intranet - is a network only for specific group of users like a company often using VPNs

Extranet - extension of intranet including also external hosts for example delivery tracking system.

1.5

client-server

C1 host send request

C2 server uses re part of resource to serve host

C3. server back in state C1. waiting.

Peer-to-peer (P2P)

a host can be both client and server and as a server is capable of forwarding request if it can not handle it.